

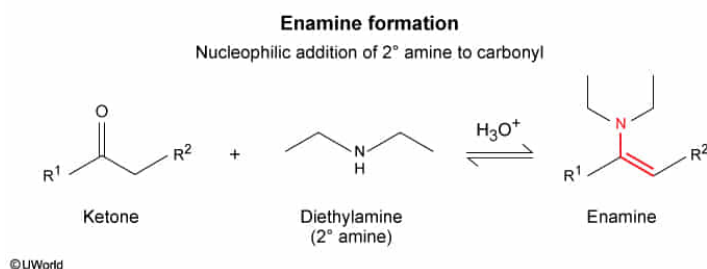
Which neutrally charged functional group forms in an acid-catalyzed reaction between a secondary amine and a ketone?

- ✔ ☒ A. Enamine [45%]
☐ B. Nitrile [5%]
☐ C. Imine [46%]
☐ D. Alkene [2%]

Correct

46% answered correctly

Explanation:



Ketones and aldehydes react with primary (1°) and secondary (2°) **amines** to form a carbon-nitrogen bond. The reaction begins with an **acid-catalyzed addition of the amine** to the carbonyl, which involves protonation of the carbonyl, nucleophilic attack of the carbonyl by the amine, and deprotonation of the amine. This forms an intermediate known as an α -aminoalcohol. The second part of the reaction is an **acid-catalyzed dehydration** involving protonation of the -OH group, loss of H_2O , and deprotonation.

The product formed is dependent on the type of amine used (1° or 2°). If a 1° amine is used, an **imine** (nitrogen analogue to ketones and aldehydes) is formed because the nitrogen has a proton available for removal in the last step, **permitting C=N bond formation**. If a **2° amine** is used, an **enamine** (nitrogen analogue to an **enol**) is formed because the nitrogen **does not have a proton available** to be removed in the last step. Therefore, a proton is removed from the α -carbon, forming a C=C bond.

(Choice B) A carbon-nitrogen bond is formed in the reaction, but the functional group is not a nitrile because the bond between carbon and nitrogen is a single bond rather than a triple bond.

(Choice C) Although a positively charged **iminium** forms as a reaction intermediate, this cation is not neutrally charged. Neutrally charged *imines* will form only if a ketone reacts with a 1° amine rather than a 2° amine.

(Choice D) Although a carbon-carbon double bond is formed in the reaction, an **alkene** indicates that the functional group contains only a double bond. Because a nitrogen atom is bonded to the sp^2 carbon, the functional group would be classified as an enamine rather than an alkene.

Educational objective:

Ketones and aldehydes undergo acid-catalyzed addition of 1° and 2° amines to form imines or enamines, respectively. Like ketones and aldehydes, imines contain a carbon-heteroatom double bond (nitrogen in imines, oxygen in ketones and aldehydes). Enamines are the nitrogen analogue to enols, containing a carbon-carbon double bond.

Time spent:0 sec

Subject:

Organic Chemistry

QID:401886

System:

Functional Groups and Their Reactions